1 1 z AB

AB C C' i

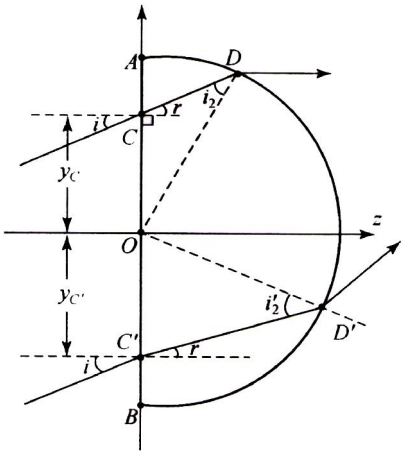
r

D D'

i_2 i'_2 $\triangle OCD$ O C

y_c

$$\frac{y_c}{\sin i_2} = \frac{R}{\sin(90° + r)}, \qquad y_c = \frac{\sin i_2}{\cos r} R . \qquad (1)$$



1

$\triangle OC'D'$ O C'

$$\frac{y_{c'}}{\sin i'_2} = \frac{R}{\sin(90° - r)}, \qquad y_{c'} = \frac{\sin i'_2}{\cos r} R . \qquad (2)$$

i r i_2 i'_2

$$i_{20} = \arcsin(1/n) \approx 41.8° \qquad (3)$$

$i_2 = i'_2 = i_{20}$ (1), (2)

$$y_{oc} = y_{oc'} = \frac{\sin i_{20}}{\cos r} R , \qquad (4)$$

$$d = 2y_{oc} = 2\frac{\sin i_{20}}{\cos r} R . \qquad (5)$$

$y_c > y_{oc}$ $y_{c'} > y_{oc'}$

i_{20} r i (4)

r 0 i 0 d

$$d_{min} = 2R\sin i_{20} = 1.33\,R . \qquad (6)$$

$i \to 90°$ $r \to 41.8°$ $r = 41.8°$

(4) d_{max} 1.79 R . (7)

2 2 φ Oz OD

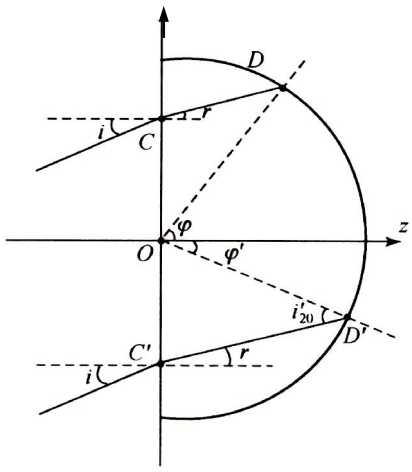
φ' Oz OD'

$$\varphi = i_{20} + r , \qquad (8)$$

$$\varphi' = i_{20} - r , \qquad (9)$$

$$\theta = \varphi + \varphi' = 2i_{20} \approx 83.6° . \qquad (10)$$

θ i θ i



2